

1) Estimating the Amount of Current Assets and Current Liabilities:

This method involves projecting the future levels of individual current assets and current liabilities based on historical data and anticipated business activities. The working capital forecast is then determined by subtracting the estimated total of current liabilities from the estimated total of current assets.

Example 22.1: ABC Manufacturing Company needs to forecast its working capital requirements for the next quarter. Based on historical data and sales projections, they estimate the following current assets and current liabilities:

Current Assets: Cash and Cash Equivalents: Rs.50,000, Accounts Receivable: Rs.120,000, Inventory: Rs.80,000, Prepaid Expenses: Rs.10,000, Total Current Assets: Rs.260,000

Current Liabilities: Accounts Payable: Rs.90,000, Short-Term Debt: Rs.30,000, Accrued Expenses: Rs.15,000, Taxes Payable: Rs.5,000, Total Current Liabilities: Rs.140,000

Working Capital Forecast = Total Current Assets – Total Current Liabilities

Working Capital Forecast = Rs.260,000 – Rs.140,000

Working Capital Forecast = Rs.120,000

2) Estimating the Cash Costs of Current Assets and Current Liabilities:

This method focuses on estimating the cash outflows related to current liabilities and the cash inflows related to current assets. It considers the timing of cash flows, providing insights into the actual cash requirements.

Example 22.2: XYZ Retail Company wants to forecast its working capital needs for the upcoming year. They estimate the following cash costs for current assets and current liabilities:

Cash Costs of Current Assets (Cash Outflows):

Cash Costs for Inventory: Rs. 500,000, Cash Costs for Prepaid Expenses: Rs. 20,000, Total Cash Costs of Current Assets: Rs. 520,000

Cash Costs of Current Liabilities (Cash Inflows):

Cash Payments for Accounts Payable: Rs.400,000, Cash Payments for Short-Term Debt: Rs.100,000, Total Cash Costs of Current Liabilities: Rs.500,000

Working Capital Forecast = Total Cash Costs of Current Assets – Total Cash Costs of Current Liabilities

Working Capital Forecast = Rs.520,000 – Rs.500,000

Working Capital Forecast = Rs.20,000

3) The Percent of Sales Method:

This approach estimates working capital needs as a percentage of sales revenue. It assumes that certain current assets and liabilities vary with sales volume.

Example 22.3: LMN Electronics Company wants to forecast its working capital requirements for the next year. Based on historical data, they find that their accounts receivable and inventory levels are typically 20% and 15% of annual sales, respectively. Their accounts payable level is typically 10% of annual sales.

Forecasted Sales for Next Year: Rs. 2,000,000

Working Capital Forecast = (Accounts Receivable + Inventory) – Accounts Payable

Working Capital Forecast = $(0.20 * \text{Rs. } 2,000,000) + (0.15 * \text{Rs. } 2,000,000) - (0.10 * \text{Rs. } 2,000,000)$

Working Capital Forecast = Rs. 400,000 + Rs. 300,000 – Rs. 200,000

Working Capital Forecast = Rs. 500,000

4) The Operating Cycle Approach:

This method estimates working capital needs based on the time it takes to convert resources into cash. It considers the time required for inventory conversion, receivables collection, and payables payment.

Example 22.4: PQR Manufacturing Company wants to forecast its working capital needs for the next six months. They estimate the following periods for their operating cycle components: Days Inventory Outstanding (DIO): 60 days

Days Sales Outstanding (DSO): 45 days

Days Payables Outstanding (DPO): 30 days

Working Capital Forecast = (DIO + DSO) – DPO

Working Capital Forecast = $(60 \text{ days} + 45 \text{ days}) - 30 \text{ days}$

Working Capital Forecast = 75 days – 30 days

Working Capital Forecast = 45 days

These methods provide different perspectives on estimating working capital needs. Businesses can use a combination of these approaches based on their specific requirements and available data to make more accurate working capital forecasts.

22.4.4 Sources of Working Capital

Working capital sources encompass the various means by which a company secures funds to meet its short-term obligations and support its day-to-day operations. These sources can be broadly categorised into short-term and long-term sources. Below are some common sources of working capital:

Short-Term Sources of Working Capital:

- a) **Trade Credit:** Companies often receive credit from suppliers, allowing them to procure goods or services and defer payment for a specified period, typically 30 to 90 days.
- b) **Bank Loans:** Short-term bank loans, such as working capital loans or overdraft facilities, offer quick access to funds to manage day-to-day operational expenses and cash flow needs.
- c) **Commercial Paper:** Creditworthy corporations may issue commercial paper, which are short-term unsecured promissory notes, to raise funds from the capital markets.
- d) **Factoring and Receivables Financing:** Companies can sell their accounts receivable to third-party factors at a discount, enabling them to obtain immediate cash against outstanding invoices.
- e) **Inventory Financing:** Inventory can be used as collateral to secure short-term loans or lines of credit.
- f) **Trade Advances:** Some businesses receive advance payments from customers for goods or services, providing an immediate infusion of working capital.

Long-Term Sources of Working Capital:

- a) **Equity Financing:** Companies can raise long-term working capital by issuing new shares of stock to investors through private placements or public offerings.
- b) **Debt Financing:** Long-term debt instruments such as corporate bonds or bank loans with longer repayment periods are used to raise working capital.
- c) **Retained Earnings:** Accumulated profits, known as retained earnings, can be reinvested in the business to serve as a long-term source of working capital.
- d) **Venture Capital and Private Equity:** Startups and high-growth firms may secure working capital from venture capital firms or private equity investors in exchange for ownership stakes.
- e) **Asset-Based Lending:** Companies can obtain long-term loans by using assets like real estate or equipment as collateral.

- f) **Debentures:** Long-term unsecured debt instruments called debentures can be issued to raise working capital from the capital markets.

22.4.5 Approaches for Determining the Financial Mix

Determining the right financing mix, or capital structure, is a crucial decision for any company, as it involves selecting the ideal proportion of debt and equity to finance its operations. The financing mix can have a significant impact on a company's risk profile, cost of capital, and overall financial performance. There are three common approaches to determining the financing mix: the hedging approach, the conservative approach, and the trade-off approach.

A) Hedging Approach:

The hedging approach aims to strike a balance between debt and equity financing to minimise the overall cost of capital while effectively managing financial risk. Companies adopting this approach calculate the optimal capital structure by minimising the weighted average cost of capital (WACC). WACC is the average cost of debt and equity, considering their respective proportions in the capital structure.

Companies following the hedging approach increase their debt levels as long as the cost of debt remains lower than the cost of equity. This strategy leads to a decrease in WACC, resulting in a lower overall cost of capital, which can potentially enhance the company's value.

However, the hedging approach recognises that excessive leverage can increase financial risk, especially if the company encounters challenges in meeting its debt obligations. Thus, there is a limit to how much debt a company should take on, and the optimal capital structure lies at the point where WACC is minimised without jeopardising the company's financial health.

B) Conservative Approach:

The conservative approach emphasises stability and financial security. Companies following this approach rely more on equity financing and maintain a lower level of debt in their capital structure. By having a higher proportion of equity, the company reduces financial risk and has a stronger ability to withstand economic downturns or unforeseen obstacles.

While the conservative approach provides a more secure financial position, it may result in a higher cost of capital compared to the hedging approach. Equity financing is often more expensive due to the costs associated with issuing new shares and the potential dilution of ownership for existing shareholders.

1) What are the methods of forecasting working capital?

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2) What are the sources of working capital?

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22.5 DIVIDEND POLICY

The establishment of a dividend policy is an important function of the finance manager. This section deals with the determination of dividend policy and the various forms of dividends.

The term dividend refers to that part of the profits of a company which is distributed amongst its shareholders. It may, therefore, be defined as the return that a shareholder gets from the company, out of its profits, on his shareholdings. According to the Institute of Chartered Accountants of India, a dividend is “a distribution to shareholders out of profits or reserves available for this purpose.”

A) Nature of Dividend Decision

The dividend decision of the firm is of crucial importance for the finance manager since it determines the amount of profit to be distributed among shareholders and the amount of profit to be required in the business (popularly termed as retained earnings) for financing its long-term growth. There is a reciprocal relationship between the cash dividends and retained earnings. Larger dividends result in less retained earnings. Fewer dividends result in larger retained earnings. While making dividend decisions, the management will consider the effect of the decision on the maximisation of shareholders' wealth. In case the payment of dividends helps the management in achieving this objective, it would be advisable to pay dividends. In case the payment of dividends does not help in achieving this objective, the management would be well advised to retain the profits and use them for financing investment programmes. Thus, the dividend decision is based on its impact on the value of the firm.

B) Conflicting Theories

There are conflicting theories regarding the impact of dividend decisions on the valuation of a firm. According to one school of thought, dividend decisions do not affect only the shareholders' wealth and so also the

valuation of the firm. According to another school of thought, dividend decision materially affects the shareholders' wealth and the valuation of the firm. For the sake of convenience of discussion, we can put the viewpoints of scholars under the following two groups:

1) Concept of Irrelevance of Dividend

This school of thought is associated with Soloman, Modigliani and Miller. According to them, dividend policy does not affect the share prices of a company and is, therefore, of no consequence. Investors do not differentiate between dividends and capital gains. Their basic desire is to earn a higher return on their investment. In case the company has adequate investment opportunities giving a higher rate of return than the cost of retained earnings, the investors would be content with the firm retaining the earnings. However, if the expected return on projects is likely to be less than what it would cost, the investors would prefer to receive the earnings (i.e. dividends). Thus, a dividend decision is a financing decision, i.e., whether to finance the company's funds requirements by retained earnings or not. In case the company has profitable investment opportunities, it will retain the earnings to finance them, otherwise distribute them. The shareholders are only interested in income whether it is in the form of dividends or capital gains.

2) Modigliani and Miller's Approach

Franco Modigliani and Merton Miller have expressed their opinion more comprehensively. They have opined that the price of shares of a firm is determined by its earning potentiality and investment policy and never by the pattern of income distribution. As observed by them, "under conditions of perfect capital markets, rational investors, absence of tax discrimination between dividend income and capital appreciation, given the firm's investment policy its dividend policy may not influence the market price of the shares."

The logic put forward by Modigliani and Miller in support of their hypothesis is that whatever increase in shareholders' wealth results from dividend payments, will be exactly offset by the effect of raising additional capital. For example, if a company, having investment opportunities, distributes all its earnings among the shareholders, it will have to raise the capital required from outside. This will increase the number of shares, resulting in a fall in the future earnings per share. Thus, whatever a shareholder has gained because of increased dividends will be neutralised completely on account of a fall in the value of shares due to a decline in the expected earnings per share.

22.6 CAPITAL BUDGETING

Capital budgeting is the process of long-term planning, encompassing proposed capital expenditures and their funding. This comprehensive approach involves both raising long-term funds and allocating them appropriately. It refers to the formal procedure a company employs to acquire

and invest in capital. Through capital budgeting, firms evaluate major fixed asset purchases, including decisions about adding, disposing, modifying, or replacing such assets.

It is essential to note that capital budgeting decisions also extend to investments in current assets that arise due to fixed asset investments. For instance, when implementing a new distribution system, the decision to invest in a new warehouse and additional inventory must be evaluated as a single package, not as separate investments.

Cases of capital budgeting decisions

Capital budgeting decisions are prevalent in various industries and organisations. Below are some cases that illustrate several types of capital budgeting decisions:

- 1) **New Product Development:** A manufacturing company is considering the launch of a new product line. The company needs to evaluate the feasibility of the project by estimating the initial investment required for production facilities, marketing costs, and expected cash flows over the next few years. They will use capital budgeting techniques like NPV, IRR, and Payback Period to assess whether the investment is financially viable and aligned with the company's strategic objectives.
- 2) **Expansion of Manufacturing Facilities:** A successful retail chain wants to expand its operations by opening new stores in different regions. The company must decide how much to invest in new store construction, inventory, and staffing. Capital budgeting analysis will help them determine the expected returns and potential risks associated with each location, allowing them to allocate resources to the most profitable expansion opportunities.
- 3) **Replacement of Machinery:** An industrial plant has ageing machinery that requires frequent maintenance, leading to increased downtime and inefficiencies. The management needs to evaluate whether investing in new, more modern equipment would improve productivity and reduce operational costs. By comparing the costs of new machinery with the expected cost savings and increased cash flows, they can make an informed capital budgeting decision.
- 4) **Mergers and Acquisitions:** A technology company is considering acquiring a smaller firm with innovative products that complement its existing offerings. Before making the acquisition, the acquirer must conduct a thorough capital budgeting analysis to determine the potential synergies, expected cash flows, and the impact of the acquisition on shareholder value.
- 5) **Research and Development (R&D) Projects:** A pharmaceutical company is deciding whether to invest in the development of a new drug. The R&D project will involve significant upfront costs and take several

years before the product is commercialised. Capital budgeting techniques, along with scenario analysis, will help assess the probability of success, potential market demand, and the overall financial viability of the R&D project.

- 6) **Capital Improvement Projects:** A municipal government needs to decide whether to invest in upgrading infrastructure, such as bridges, roads, or water treatment plants. Capital budgeting analysis will consider the initial investment, the estimated life of the improvement, maintenance costs, and the benefits to the community, including potential economic growth and increased property values.

22.6.1 Importance of Capital Budgeting

Capital budgeting holds immense significance for businesses and organisations due to various compelling reasons:

- 1) **Long-term investment choices:** Capital budgeting involves making decisions about long-term investments that have enduring effects. These investments typically require substantial financial resources, and their success or failure can profoundly impact the company's future growth and profitability.
- 2) **Optimal resource allocation:** By utilising capital budgeting techniques, companies can efficiently allocate their limited resources, such as funds and workforce, to projects that are expected to yield the highest returns. This ensures that resources are utilised effectively to create maximum value for the organisation.
- 3) **Maximising shareholder wealth:** The main goal of capital budgeting is to pinpoint and choose projects that contribute to the growth of the company's shareholders' wealth. By investing in projects with favourable net present values (NPVs) or significant internal rates of return (IRRs), the company can boost shareholder value and appeal to potential investors.
- 4) **Strategic alignment:** Capital budgeting decisions are closely tied to a company's strategic objectives. It ensures that projects chosen for investment align with the organisation's long-term goals, enabling the allocation of resources to endeavours that contribute to the company's overall vision.
- 5) **Risk assessment:** Capital budgeting involves a comprehensive evaluation of the risks associated with different investment projects. Understanding potential risks and uncertainties enables companies to make informed decisions and implement risk mitigation strategies when necessary.
- 6) **Capital rationing:** In situations where capital is limited, capital budgeting helps prioritise investment opportunities based on their

expected returns and strategic importance. It enables companies to optimise the use of available funds by selecting the most promising projects.

- 7) **Efficient use of funds:** Capital budgeting prevents investments in projects with negative NPVs or low returns, thus avoiding wasteful use of financial resources. Focusing on profitable projects allows companies to utilise their funds efficiently to achieve growth and success.

22.6.2 Kinds of Capital Investment Proposals

A firm may be presented with multiple investment proposals, and their decision to adopt one, some, or all of them depends on their categorisation as independent, contingent, or mutually exclusive.

a) Independent Proposals:

These investment proposals are non-competing, implying that accepting one proposal does not prevent the acceptance of another. The firm can directly assess each proposal's return on investment and accept those that offer returns higher than the desired rate of return.

Proposals meeting this criterion are accepted, while the rest are rejected.

b) Contingent or Dependent Proposals:

These proposals' acceptance relies on the approval of one or more other proposals. For instance, the acquisition of a new machine may be contingent upon a substantial plant expansion. In such cases, the dependent investment proposal should be presented along with the related proposal it relies on, providing a comprehensive perspective.

c) Mutually Exclusive Proposals:

These proposals are competing in such a way that accepting one automatically excludes the acceptance of others. **For instance**, if a company is evaluating two temperature control systems and decides to invest in one system, it means rejecting the other. Consequently, the firm needs to employ a decision-making technique to choose the most favourable option, leading to the exclusion of other alternatives.

Check Your Progress 4

- 1) What are the theories regarding the impact of dividend policy?

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- 2) What is capital budgeting? Discuss what is its importance?

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22.6.3 Factors Affecting Capital Investment Decision

Factors affecting capital investment decisions can vary depending on the industry, company, and specific project. Below are some common factors that influence such decisions:

- 1) **Market Demand:** Market demand for the product or service being offered is a fundamental factor influencing capital investment decisions. Companies need to assess the potential size of the target market, the growth rate, and the demand forecast to determine the potential sales volume and revenue generation. **Example:** Regarding the smartphone project, a comprehensive market analysis suggests that the smartphone market is projected to experience an annual growth rate of 8%, and the company's product is anticipated to secure a 10% market share.
- 2) **Competition:** The level of competition in the market is a critical consideration. A highly competitive market may lead to price wars and reduced profit margins, affecting the project's financial feasibility. **Example:** The smartphone market has several well-established brands with strong customer loyalty, making it a highly competitive space for new entrants.
- 3) **Cost of Capital:** The cost of capital signifies the return that investors anticipate for funding an investment. It plays a crucial role in establishing the discount rate utilised to assess future cash flows. **Example:** If the company's cost of capital is 12%, it implies that any investment must generate returns exceeding 12% to be deemed financially feasible.
- 4) **Initial Investment Cost:** The total upfront cost required to start the project, including machinery, equipment, land, and working capital, needs to be carefully evaluated. **Example:** The initial investment for the smartphone project includes ₹7,00,00,000 for machinery, ₹2,00,00,000 for land, and ₹1,00,00,000 for working capital.
- 5) **Cash Flows:** Projected cash inflows and outflows over the life of the project are essential for evaluating profitability. Companies must estimate expected revenue and operating expenses. **Example:** The smartphone project expects ₹20,00,00,000 in annual sales revenue and annual operating expenses of ₹12,00,00,000.
- 6) **Payback Period:** The payback period is a measure of the time required to recoup the initial investment. Typically, a shorter payback period is